

GENERAL KNOWLEDGE

- $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

- $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$

- $S_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = t_{xy} \cdot S_x \cdot S_y$

- $S_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

- $S_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$

- $\sqrt{\text{VAR}(\hat{\beta}_i)} = \text{SE}(\hat{\beta}_i)$

SUM OF RESIDUALS $\rightarrow$ trick to remember: $y - \hat{y}$

$$\sum e_i^2 = \text{SSE} = \sum (y_i - \hat{y}_i)^2 = \sum (y_i - \bar{y})^2 - \hat{\beta}_1^2 \sum (x_i - \bar{x})^2$$

- $\hat{\sigma}^2 = \frac{1}{n} \sum e_i^2 = \frac{1}{n} \text{SSE} \rightarrow \text{BIASED ESTIMATE OF THE VARIANCE}$

- $S^2 = \frac{1}{n-p} \sum e_i^2 = \frac{n}{n-p} \hat{\sigma}^2 \rightarrow \text{UNBIASED ESTIMATE OF THE VARIANCE}$
 $\hookrightarrow \frac{\text{SSE}}{n-p} = \frac{1}{n-p} \sum (y_i - \hat{y}_i)^2$

- $\bullet$ SAMPLE SPACE: $y = \mathbb{R}^n$ $\downarrow$ SAMPLE SIZE
- $\bullet$ PARAMETER SPACE: $\theta = \mathbb{R}^p \times \underbrace{(0, \infty)}_{\text{IR}^+ \text{ (variance)}}$

- $\bullet$ RESIDUAL STANDARD ERROR: $\sqrt{\frac{1}{n-p} \sum (y_i - \hat{y}_i)^2}$
 $\hookrightarrow$ CAN BE USED TO FIND SSE

$$\text{SST} = \sum (y_i - \bar{y})^2$$

$$\text{SSE} = \sum (y_i - \hat{y}_i)^2$$

$$\text{SSR} = \sum (\hat{y}_i - \bar{y})^2$$

$$\text{SST} = \text{SSE} + \text{SSR}$$

$$R^2 = \frac{\text{SSR}}{\text{SST}} \quad t_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{\frac{1}{n-1} \sum (y_i - \bar{y})(x_i - \bar{x})}{\sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2} \sqrt{\frac{1}{n-1} \sum (y_i - \bar{y})^2}}$$



R -ADJUSTED $R^2_{\text{ADJ}} = 1 - \left[(1 - R^2) \cdot \frac{(n-1)}{(n-p)} \right]$

Hypothesis testing

T-test

FOR INFERENCE ON β

$$\begin{cases} H_0: \beta_1 = 0 \\ H_1: \beta_1 \neq 0 \end{cases} \quad e_i^{\text{OBS}} \sim t_{n-p}$$

$$e_i^{\text{OBS}} = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{VAR}(\hat{\beta}_1)}} = \frac{\hat{\beta}_1 - \beta_1}{\text{SE}(\hat{\beta}_1)}$$

$$\alpha^{\text{OBS}} = P(|t_{n-p}| \geq |e_i^{\text{OBS}}|) = 2 \min \{ P(t_{n-p} \leq -e_i^{\text{OBS}}), P(t_{n-p} \geq e_i^{\text{OBS}}) \}$$

$$\alpha^{\text{OBS}} = 2(1 - \sim) \quad \begin{matrix} \alpha^{\text{OBS}} & \alpha & \alpha^{\text{OBS}} \\ \text{R. NO } R_1 & & \end{matrix}$$

find α value

$$t_{n-p; \sim} = e_i^{\text{OBS}} \quad \leftarrow \text{COMPUTED BEFORE}$$

$\hookrightarrow$ TO FIND α^{OBS} YOU NEED TO DO THIS VALUE AND DO 1 - $\alpha^{\text{OBS}} = 1 - \sim$

$\triangle$ IF TWO TAILS REMEMBER TO MULTIPLY BY 2 $\alpha^{\text{OBS}} = 2(1 - \sim)$

OTHER CASES:

- $\begin{cases} H_0: \beta_2 = -1 \\ H_1: \beta_2 < -1 \end{cases} \quad Z_{\text{OBS}} \sim N(0,1)$
- $Z_{\text{OBS}} = \frac{\hat{\beta}_2 - (-1)}{\sqrt{\text{VAR}(\hat{\beta}_2)}} \quad \alpha^{\text{OBS}} = 1 - \Phi(|Z_{\text{OBS}}|)$

- $\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 < 0 \end{cases} \quad Z_{\text{OBS}} \sim N(0,1)$
- $Z_{\text{OBS}} = \frac{\hat{\beta}_3 - 0}{\text{SE}(\hat{\beta}_3)} \quad \alpha^{\text{OBS}} = \Phi(|Z_{\text{OBS}}|)$

$\bullet$ FIND CORRECT $n-p$ ROW

$\bullet$ FIND IN THE TABLE t_{OBS}

$\bullet$ GO UP TO THE COLUMN

NAME: THAT IS $1 - \alpha^{\text{OBS}}$

$\bullet$ REMEMBER TO MULTIPLY

BY 2 IF TWO TAILS

			p					
			0.90	0.95	0.975	0.99	0.995	0.9975
standard Normal	z_p		1.2816	1.6449	1.9600	2.3263	2.5758	2.8070
t with 21 df	$t_{21,p}$		1.3232	1.7207	2.0796	2.5176	2.8314	3.1352
t with 22 df	$t_{22,p}$		1.3212	1.7171	2.0739	2.5083	2.8188	3.1188
t with 23 df	$t_{23,p}$		1.3195	1.7139	2.0687	2.4999	2.8073	3.1040
t with 31 df	$t_{31,p}$		1.3095	1.6955	2.0395	2.4528	2.7440	3.0221
t with 32 df	$t_{32,p}$		1.3086	1.6939	2.0369	2.4487	2.7385	3.0149
t with 33 df	$t_{33,p}$		1.3077	1.6924	2.0345	2.4448	2.7333	3.0082

F-test

FOR INFERENCE ON R^2 and test significance of overall model

$$\begin{cases} H_0: R^2 = 0 \\ H_1: R^2 > 0 \end{cases} \quad F_{\text{OBS}} \sim F_{p-1; n-p}$$

$$F_{\text{OBS}} = \frac{\frac{\text{SSR}}{p-1}}{\frac{\text{SSE}}{n-p}} = \frac{R^2}{1-R^2} \cdot \frac{(n-p)}{(p-1)} = \frac{\text{SSR}}{\text{SSE}} \cdot \frac{(n-p)}{(p-1)} = F_{\text{OBS}}^2$$

$$\alpha^{\text{OBS}} = P(|F| \geq |F_{\text{OBS}}|) \quad \alpha^{\text{OBS}} = 1 - \sim$$

$$\begin{matrix} \alpha^{\text{OBS}} & \alpha & \alpha^{\text{OBS}} \\ \text{R. NO } R_1 & & \end{matrix}$$

find α value

$$f_{p-1, n-p; \sim} = f_{\text{OBS}}$$

$\hookrightarrow$ TO FIND α^{OBS} YOU NEED TO DO THIS VALUE AND DO 1 - $\alpha^{\text{OBS}} = 1 - \sim$

$\bullet$ FIND CORRECT $p-1$

$\bullet$ AND $n-p$ ROW

$\bullet$ FIND IN THE TABLE f_{OBS}

$\bullet$ GO UP TO THE COLUMN

NAME: THAT IS $1 - \alpha^{\text{OBS}}$

			p					
			0.90	0.95	0.975	0.99	0.995	0.9975
$f_{1,n-p}$			2.0472	2.5977	3.0932	3.7102	4.2501	4.8366
$f_{2,n-p}$			1.9849	2.4422	2.9023	3.5200	4.0469	4.5807
$f_{3,n-p}$			1.9531	2.3748	2.8077	3.4057	3.8822	4.3826
$f_{4,n-p}$			1.9319	2.3201	2.7313	3.2986	3.7502	4.2343
$f_{5,n-p}$			1.9068	2.2691	2.6556	3.1929	3.6886	4.1653
$f_{6,n-p}$			1.9132	2.2127	2.5750	3.0983	3.6019	4.1185
$f_{7,n-p}$			1.8702	2.2444	2.6202	3.1267	3.5210	3.9271
$f_{8,n-p}$			1.8348	2.1888	2.5434	3.0508	3.3919	3.7788
$f_{9,n-p}$			2.8223	3.8486	5.1284	7.3309	9.4922	12.2271
$f_{10,n-p}$			2.5796	3.4179	4.4263	6.0021	7.6688	9.5865
$f_{11,n-p}$			2.4886	3.1229	3.9587	5.2967	6.5200	7.9832
$f_{12,n-p}$			2.2816	2.9884	3.6527	4.7463	5.7516	6.9197
$f_{13,n-p}$			2.7953	3.7998	5.0521	7.3073	9.2290	11.9983
$f_{14,n-p}$			2.5504	3.3670	4.3491	5.9712	7.2066	9.3740
$f_{15,n-p}$			2.5777	3.0703	3.8806	5.1776	6.3691	7.7816
$f_{16,n-p}$			2.2491	2.8543	3.5468	4.6282	5.5984	6.7255

Confidence interval

(α): $1 - \alpha = 0.95$

$$CI(1-\alpha) = \left(\hat{\beta}_1 - t_{n-p; 1-\alpha/2} \sqrt{\text{VAR}(\hat{\beta}_1)}; \hat{\beta}_1 + t_{n-p; 1-\alpha/2} \sqrt{\text{VAR}(\hat{\beta}_1)} \right)$$

SIMPLE LINEAR MODELS

ASSUMPTIONS

- ABSENCE OF SYSTEMATIC ERROR $\rightarrow E[\epsilon_i] = 0 \rightarrow E[y_i] = \beta_1 + \beta_2 x_i$ (LINEARITY)
- HOMOGENEITY OF THE ERRORS $\rightarrow \text{VAR}(\epsilon_i) = \sigma^2 > 0 \rightarrow \text{VAR}(y_i) = \sigma^2 > 0$ (HOMOGENEITY OF THE RESPONSE)
- UNCORRELATED ERRORS $\rightarrow \text{COV}(\epsilon_i, \epsilon_k) = 0$ FOR $i \neq k \rightarrow \text{COV}(y_i, y_k) = 0$ FOR $i \neq k$
- $x_1, \dots, x_n$ FIXED AND NON-STOCHASTIC
- x_i CAN NOT ALL BE EQUAL

$$\begin{aligned} \bullet \hat{\beta}_1 &= \bar{y} - \hat{\beta}_2 \bar{x} \\ \bullet \hat{\beta}_2 &= \frac{s_{xy}}{s_{x^2}} = \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} = \frac{-n \bar{x} \bar{y} + \sum_{i=1}^n x_i y_i}{-n \bar{x}^2 + \sum_{i=1}^n x_i^2} = \frac{\text{COV}(x, y)}{\text{DEV}(x)} \end{aligned}$$

$$R^2 = r_{xy}^2 \quad \text{! SIMPLE LINEAR REGRESSION MODEL ONLY}$$

$$\begin{aligned} \bullet \hat{\text{VAR}}(\hat{\beta}_1) &= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) \\ &\quad \text{CAN USE } s^2 \\ \bullet \hat{\text{VAR}}(\hat{\beta}_2) &= \frac{s^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \end{aligned}$$

point estimate and confidence interval of a new observation

POINT ESTIMATE

$$\text{NEW DATA } x^* = s \rightarrow y^* = \hat{\beta}_1 + \hat{\beta}_2 x^*$$

CONFIDENCE INTERVAL

$$\begin{aligned} \hat{y}^* &\sim N \left[\underbrace{\hat{\beta}_1 + \hat{\beta}_2 x^*}_{\hat{y}^*}; s^2 \left(\frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) \right] \\ \text{CI: } &\left\{ \hat{y}^* - t_{n-2; 1-\alpha/2} \sqrt{s^2 \left(\frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)}; \hat{y}^* + t_{n-2; 1-\alpha/2} \sqrt{s^2 \left(\frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)} \right\} \end{aligned}$$

SIMPLE GAUSSIAN LINEAR MODELS

ASSUMPTIONS

- ABSENCE OF SYSTEMATIC ERROR $\rightarrow E[\epsilon_i] = 0 \rightarrow E[y_i] = \beta_1 + \beta_2 x_i$ (LINEARITY)
- HOMOGENEITY OF THE ERRORS $\rightarrow \text{VAR}(\epsilon_i) = \sigma^2 > 0 \rightarrow \text{VAR}(y_i) = \sigma^2 > 0$ (HOMOGENEITY OF THE RESPONSE)
- UNCORRELATED ERRORS $\rightarrow \text{COV}(\epsilon_i, \epsilon_k) = 0$ FOR $i \neq k \rightarrow \text{COV}(y_i, y_k) = 0$ FOR $i \neq k$

$$\bullet \text{GAUSSIAN DISTRIBUTION OF THE ERRORS} \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

SAME OF SIMPLE LINEAR MODELS

SAME OF SIMPLE LINEAR MODELS

- $x_1, \dots, x_n$ FIXED AND NON-STOCHASTIC
- x_i CAN NOT ALL BE EQUAL

MULTIPLE LINEAR REGRESSION MODELS

ASSUMPTIONS:

- ABSENCE OF SYSTEMATIC ERROR $\rightarrow E[\epsilon_i] = 0 \rightarrow E[y_i] = \beta_1 + \beta_2 x_i$ (LINEARITY)
- HOMOGENEITY OF THE ERRORS $\rightarrow \text{VAR}(\epsilon_i) = \sigma^2 > 0 \rightarrow \text{VAR}(y_i) = \sigma^2 > 0$ (HOMOGENEITY OF THE RESPONSE)
- UNCORRELATED ERRORS $\rightarrow \text{COV}(\epsilon_i, \epsilon_k) = 0$ FOR $i \neq k \rightarrow \text{COV}(y_i, y_k) = 0$ FOR $i \neq k$
- LINEARITY: CONDITIONALLY $\mathcal{Y}_i = \beta_1 + \beta_2 x_i$
- $\epsilon_i \sim N(0, \sigma^2) \quad i = 1, \dots, n$
- THE VECTORS $x_i \in \mathbb{R}^n \quad i = 1, \dots, n$ ARE LINEARLY INDEPENDENT $\rightarrow \text{RANK}(X) = n$ (FULL RANK)

$$y_i = \beta_1 + \beta_2 x_{i1} + \beta_3 x_{i2} + \epsilon_i \quad \text{! FOLLOW THE ORDER}$$

$$\text{BOOLEAN DUMMY VARIABLE} \quad D_1 = \begin{cases} 1 & \text{MALE} \\ 0 & \text{FEMALE} \end{cases} \quad \text{! FOLLOW THE ORDER}$$

$$\text{TERNARY DUMMY VARIABLE (SPUT IN 2)} \quad D_1 = \begin{cases} 1 & \text{IF A} \\ 0 & \text{OTHERWISE} \end{cases} \quad D_2 = \begin{cases} 1 & \text{IF B} \\ 0 & \text{OTHERWISE} \end{cases} \quad \text{! FOLLOW THE ORDER}$$

$$y = X\beta + \epsilon$$

$$\begin{matrix} y & X & \beta & \epsilon \\ (n \times 1) & (n \times p) & (p \times 1) & (n \times 1) \end{matrix}$$

- $y \sim N_p(y; \beta; I_p)$
- $X = \begin{bmatrix} x_1 & x_2 & x_3 & \dots & x_p \end{bmatrix}$
- $\epsilon \sim N_p(0; \sigma^2 I_p)$
- $\text{RANK}(X) = p$ (FULL RANK)

$$(X^T X)^{-1} = \begin{bmatrix} c_{1,1} & c_{1,2} & c_{1,3} \\ c_{2,1} & c_{2,2} & c_{2,3} \\ c_{3,1} & c_{3,2} & c_{3,3} \end{bmatrix} \quad (p \times p)$$

$$\text{CAN USE } S^2 = \frac{SSE}{n-p}$$

$$\text{COV}(\hat{\beta}_1, \hat{\beta}_2) = \sigma^2 c_{12}$$

$$c_{12} = \rho(c_{11}, c_{22})^{1/2} = \rho \frac{\sqrt{\text{VAR}(\hat{\beta}_1)} \sqrt{\text{VAR}(\hat{\beta}_2)}}{S^2}$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$\begin{matrix} \text{ESTIMATE OF } \beta \\ (p \times 1) = (p \times p) (p \times 1) \end{matrix} \quad X^T y = \begin{bmatrix} - \\ - \\ - \end{bmatrix} \quad (p \times 1)$$

VARIANCE-COVARIANCE MATRIX

$$\text{VAR}(\hat{\beta}) = \begin{bmatrix} \text{VAR}(\hat{\beta}_1) & \text{COV}(\hat{\beta}_1, \hat{\beta}_2) & \text{COV}(\hat{\beta}_1, \hat{\beta}_3) \\ \text{COV}(\hat{\beta}_2, \hat{\beta}_1) & \text{VAR}(\hat{\beta}_2) & \text{COV}(\hat{\beta}_2, \hat{\beta}_3) \\ \text{COV}(\hat{\beta}_3, \hat{\beta}_1) & \text{COV}(\hat{\beta}_3, \hat{\beta}_2) & \text{VAR}(\hat{\beta}_3) \end{bmatrix} \quad (p \times p)$$

$$\text{CAN USE } S^2 = \frac{SSE}{n-p}$$

$$= \sigma^2 (X^T X)^{-1}$$

$$\text{CAN USE } S^2 = \frac{SSE}{n-p}$$

$$\text{COV}(\hat{\beta}_1, \hat{\beta}_2) = \sigma^2 c_{12}$$

Distribution of b2 - b3

$$\hat{\beta}_2 - \hat{\beta}_3 \sim N \left[\beta_2 - \beta_3; \text{VAR}(\hat{\beta}_2) + \text{VAR}(\hat{\beta}_3) - 2\text{COV}(\hat{\beta}_2, \hat{\beta}_3) \right]$$

$$\text{VAR}(\hat{\beta}_2) = \frac{c_{22}}{p(\hat{\beta}_1, \hat{\beta}_3)} \cdot S^2$$

$$\text{VAR}(\hat{\beta}_3) = \frac{c_{33}}{p(\hat{\beta}_1, \hat{\beta}_2)} \cdot S^2$$

$$\text{COV}(\hat{\beta}_2, \hat{\beta}_3) = \text{COV}(\hat{\beta}_2, \hat{\beta}_3) = \frac{c_{23}}{p(\hat{\beta}_1, \hat{\beta}_3)} \cdot S^2$$

Hypothesis b2 = b3

$$\begin{cases} H_0: \beta_2 = \beta_3 \\ H_1: \beta_2 \neq \beta_3 \end{cases} \rightarrow \begin{cases} H_0: \beta_2 - \beta_3 = 0 \\ H_1: \beta_2 - \beta_3 \neq 0 \end{cases}$$

$$t_{23}^{OBS} \sim t_{n-p} \quad t_{23}^{OBS} = \frac{\hat{\beta}_2 - \hat{\beta}_3 - 0}{\sqrt{\text{VAR}(\hat{\beta}_2) + \text{VAR}(\hat{\beta}_3) - 2\text{COV}(\hat{\beta}_2, \hat{\beta}_3)}}$$

$$\alpha^{OBS} = P(|t_{n-p}| \geq |t_{23}^{OBS}|) = 2 \min \{ P(t_{n-p} < -t_{23}^{OBS}); P(t_{n-p} > t_{23}^{OBS}) \}$$

Compare models

- MOD 1: DEFAULT MODEL $\rightarrow y_i = \beta_1 + \beta_2 x_{i1} + \beta_3 x_{i2} + \beta_4 D_{1i} + \beta_5 D_{2i}$
- MOD 2: MOD 1 $\pm$ COVARIATES $\rightarrow y_i = \beta_1 + \beta_2 x_{i1} + \beta_3 x_{i2} + \beta_4 D_{1i} + \beta_5 D_{2i} + \beta_6 D_{1i} x_{i1}$

$$\text{TEST STATISTIC} \quad \begin{cases} H_0: \beta_6 = 0 \\ H_1: \beta_6 \neq 0 \end{cases}$$

$$F_{OBS} = \frac{\frac{SSE(1) - SSE(2)}{p - p_0}}{\frac{SSE(2)}{n - p}} \sim F_{p-p_0, n-p}$$

(piccolo mod) - (grande mod)

(piccolo mod) $\rightarrow$ Mod 1

(grande mod) $\rightarrow$ Mod 2

- Nei nested models SST è uguale per entrambi i models
- SSE di un modello con solo intercept (b1) è uguale a SST

- Se il comparison è tra un modello e uno con solo l'intercept (b1) allora posso usare la formula:

$$F_{OBS} = \frac{\frac{SSR}{p-1}}{\frac{SSE}{n-p}} = \frac{R^2}{1-R^2} \frac{(n-p)}{(p-1)} = \frac{SSR}{SSE} \frac{(n-p)}{(p-1)} = T_{OBS}^2 \sim F_{p-1, n-p}$$

- reject (h0) => modello con più covariates è meglio
- not reject (h0) => modello con solo intercept è meglio

GENERALISED LINEAR MODEL

OUTPUT

Boolean (binary)

PROBIT LINK FUNCTION

ASSUMPTION
• $y_i \sim \text{Bernoulli}(\pi_i)$
• $\pi_i = \Phi(\beta_0 + \beta_1 x_i)$
• $\Phi(\pi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\pi} e^{-t^2/2} dt$

LOGIT LINK FUNCTION

ASSUMPTION
• $y_i \sim \text{Bernoulli}(\pi_i)$
• $\pi_i = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$
• $\pi(\pi) = \text{logit}\left(\frac{\pi}{1-\pi}\right)$

Count (number)

POISSON REGRESSION

ASSUMPTION
• $y_i \sim \text{Poisson}(\mu_i)$ with $\mu_i = e^{\beta_0 + \beta_1 x_i}$
• $\pi(\pi) = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$
• $\pi(\pi) = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$

LOGIT: ODDS SCALE & ODDS RATIO OF A BOOLEAN VARIABLE D_i

LOGIT LINK FUNCTION

EX: $\text{logit}\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 D_i \rightarrow \pi_i = \frac{e^{\beta_0 + \beta_1 D_i}}{1 + e^{\beta_0 + \beta_1 D_i}}$
"Odds"
• $\pi_i | D_i = 1 = \frac{e^{\beta_0 + \beta_1}}{1 + e^{\beta_0 + \beta_1}}$
• $\pi_i | D_i = 0 = \frac{e^{\beta_0}}{1 + e^{\beta_0}}$
• $\pi_i | D_i = 1 - \pi_i | D_i = 0 = \frac{e^{\beta_0 + \beta_1}}{e^{\beta_0}} = e^{\beta_1}$
• $\pi_i | D_i = 1 - \pi_i | D_i = 0 = \frac{e^{\beta_0 + \beta_1}}{e^{\beta_0}} = e^{\beta_1}$

ODDS SCALE

raggruppato in base a X_i , con $Y=1$ (pos) sopra
• per $D_i = 1$: $\frac{e^{\beta_0 + \beta_1}}{1 + e^{\beta_0 + \beta_1}}$
• per $D_i = 0$: $\frac{e^{\beta_0}}{1 + e^{\beta_0}}$
• per $D_i = 1$: $\frac{e^{\beta_0 + \beta_1}}{e^{\beta_0}} = e^{\beta_1}$

When $D_i = 1$ expected <value of e^{β_1} > *100> success every 100 failure
When $D_i = 0$ expected <value of e^{β_1} > *100> success every 100 failure

ODDS RATIO

can D_i-typed sopra

$\frac{\pi_i | D_i = 1}{\pi_i | D_i = 0} = \frac{e^{\beta_0 + \beta_1}}{e^{\beta_0}} = e^{\beta_1}$
 $\frac{\pi_i | D_i = 1}{\pi_i | D_i = 0} = \frac{e^{\beta_0 + \beta_1}}{e^{\beta_0}} = e^{\beta_1}$

INTERPRETATION

The odds for the group $D_i = 1$ are multiplied by <value> to obtain the odds for the group $D_i = 0$

LOGIT: ODDS RATIO OF A CONTINUOUS VARIABLE X_i

OVERCONFIDENCE X_i (β_1)

odds
• NO X_i : $\text{logit}\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 X_i + \beta_2 D_i$
• YES X_i : $\text{logit}\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 (x_i + 1) + \beta_2 D_i$

odds ratio

$\frac{\pi_i | x_i + 1}{\pi_i | x_i} = \frac{e^{\beta_0 + \beta_1 (x_i + 1) + \beta_2 D_i}}{e^{\beta_0 + \beta_1 x_i + \beta_2 D_i}} = e^{\beta_1}$
 $\frac{\pi_i | x_i + 1}{\pi_i | x_i} = \frac{e^{\beta_0 + \beta_1 (x_i + 1) + \beta_2 D_i}}{e^{\beta_0 + \beta_1 x_i + \beta_2 D_i}} = e^{\beta_1}$

$e^{\beta_1} = e^{0.476} \approx 1.61$ => an increase of one unit in the overconfidence lead to an increase in the odds of the odds of 1.61

=> an increase of one unit in the overconfidence lead to a change in the log-odds of β_1 , while keeping the other covariates fixed.

=> an increase of 1 unit in the overconfidence lead to a change/multiplication of the initial odds(at x_0) of 1.61

DOMINANCE X_i (β_2)

odds
• NO X_i : $\text{logit}\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 X_i + \beta_2 D_i$
• YES X_i : $\text{logit}\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 X_i + \beta_2 (D_i + 1)$

odds ratio

$\frac{\pi_i | D_i + 1}{\pi_i | D_i} = \frac{e^{\beta_0 + \beta_1 X_i + \beta_2 (D_i + 1)}}{e^{\beta_0 + \beta_1 X_i + \beta_2 D_i}} = e^{\beta_2}$
 $\frac{\pi_i | D_i + 1}{\pi_i | D_i} = \frac{e^{\beta_0 + \beta_1 X_i + \beta_2 (D_i + 1)}}{e^{\beta_0 + \beta_1 X_i + \beta_2 D_i}} = e^{\beta_2}$

$e^{\beta_2} = e^{0.673} \approx 1.96$ => an increase of one unit in the dominance lead to an increase of 4.76 in the odds of the odds

=> an increase of one unit in the dominance lead to a change in the log-odds of β_2 , while keeping the other covariates fixed.

=> an increase of 1 unit in the dominance lead to a change/multiplication of the initial odds(at x_0) of 4.76

HYPOTESIS TESTING

$$\left\{ \begin{array}{l} H_0: \beta_1 = 0 \\ H_1: \beta_1 \neq 0 \end{array} \right. \quad Z_{\text{test}} = \frac{\hat{\beta}_1}{\sqrt{\text{var}(\hat{\beta}_1)}} \quad Z_{\text{test}} = \frac{\hat{\beta}_1}{\sqrt{\text{var}(\hat{\beta}_1)}} \quad Z_{\text{test}} = \frac{\hat{\beta}_1}{\sqrt{\text{var}(\hat{\beta}_1)}}$$

$$\alpha^{0.05} = P(|Z| > |Z_{\alpha^{0.05}}|) = 2 \cdot (1 - \Phi(|Z_{\alpha^{0.05}}|)) \quad \frac{\alpha^{0.05}}{2} = \alpha^{0.025}$$

CONFIDENCE INTERVAL

(CI for odds ratio)

$$\left(e^{\hat{\beta}_1 - 1.96 \sqrt{\text{var}(\hat{\beta}_1)}}, e^{\hat{\beta}_1 + 1.96 \sqrt{\text{var}(\hat{\beta}_1)}} \right)$$

(confidence interval)

$$CI(1 - \alpha) = \left[\hat{\beta}_1 - Z_{\alpha/2} \sqrt{\text{var}(\hat{\beta}_1)}, \hat{\beta}_1 + Z_{\alpha/2} \sqrt{\text{var}(\hat{\beta}_1)} \right]$$

LIKELIHOOD RATIO TEST

$\chi^2_{(k-1)}$ < observed at 5% tail
 $\chi^2_{(k-1)}$ > obs. < observed at 5% tail

• saturated model: $\mu = \mu_{ijk}$ (no constraints)
• restricted model: $\mu = \mu_{ij}$ (constraints)
• nested model: $\mu = \mu_{ij}$ (constraints)

LIKELIHOOD RATIO TEST

• saturated model: $\mu = \mu_{ijk}$ (no constraints)
• restricted model: $\mu = \mu_{ij}$ (constraints)
• nested model: $\mu = \mu_{ij}$ (constraints)

TEST OVERALL SIGNIFICANCE

$$\chi^2_{(k-1)} = 2 \ln \left(\frac{L(\mu_{ijk})}{L(\mu_{ij})} \right) = 2 \left\{ \sum_{i,j,k} \ln \left(\frac{\mu_{ijk}}{\mu_{ij}} \right) - \sum_{i,j,k} \ln \left(\frac{\mu_{ij}}{\mu_{ij}} \right) \right\} = 2 \left\{ \sum_{i,j,k} \ln \left(\frac{\mu_{ijk}}{\mu_{ij}} \right) - \sum_{i,j,k} \ln \left(\frac{\mu_{ij}}{\mu_{ij}} \right) \right\}$$

$$\chi^2_{(k-1)} = 2 \ln \left(\frac{L(\mu_{ijk})}{L(\mu_{ij})} \right) = 2 \left\{ \sum_{i,j,k} \ln \left(\frac{\mu_{ijk}}{\mu_{ij}} \right) - \sum_{i,j,k} \ln \left(\frac{\mu_{ij}}{\mu_{ij}} \right) \right\}$$

$$\chi^2_{(k-1)} = 2 \ln \left(\frac{L(\mu_{ijk})}{L(\mu_{ij})} \right) = 2 \left\{ \sum_{i,j,k} \ln \left(\frac{\mu_{ijk}}{\mu_{ij}} \right) - \sum_{i,j,k} \ln \left(\frac{\mu_{ij}}{\mu_{ij}} \right) \right\}$$

$$\chi^2_{(k-1)} = 2 \ln \left(\frac{L(\mu_{ijk})}{L(\mu_{ij})} \right) = 2 \left\{ \sum_{i,j,k} \ln \left(\frac{\mu_{ijk}}{\mu_{ij}} \right) - \sum_{i,j,k} \ln \left(\frac{\mu_{ij}}{\mu_{ij}} \right) \right\}$$

POISSON

• $I(\mu) \rightarrow \sum_{i=1}^n y_i \ln \mu_i - \mu_i$ (use of Poisson distribution)
• $I(\mu) \rightarrow \sum_{i=1}^n y_i \ln \mu_i - \mu_i$ (use of Poisson distribution)
• $I(\mu) \rightarrow \sum_{i=1}^n y_i \ln \mu_i - \mu_i$ (use of Poisson distribution)

LOGIT

• $I(\mu) \rightarrow \sum_{i=1}^n y_i \ln \mu_i - \mu_i$ (use of Poisson distribution)
• $I(\mu) \rightarrow \sum_{i=1}^n y_i \ln \mu_i - \mu_i$ (use of Poisson distribution)
• $I(\mu) \rightarrow \sum_{i=1}^n y_i \ln \mu_i - \mu_i$ (use of Poisson distribution)

"Null deviance" and "Residual deviance" explained

a) the residual deviance is the dev. made b/w the structural model and the proposed model:

$$R(\mu) = \sum_{i=1}^n \ln \left(\frac{\mu_i}{\mu_i} \right) = \sum_{i=1}^n \ln \left(\frac{\mu_i}{\mu_i} \right)$$

where $R(\mu)$ is the maximum of the log-likelihood under a model with n parameters.

The null deviance is $R(\mu_0) = \sum_{i=1}^n \ln \left(\frac{\mu_i}{\mu_i} \right) = \sum_{i=1}^n \ln \left(\frac{\mu_i}{\mu_i} \right)$

where $R(\mu_0)$ is the maximum of the log-likelihood under a model with a single parameter, μ .

GOODNESS OF THE FIT

$$D(\mu_0) = 2 \left\{ \ln \left(\frac{L(\mu)}{L(\mu_0)} \right) \right\}$$

• $D(\mu_0) < 0$ (model) => model not good
• $D(\mu_0) > 0$ (model) => model good

$$[0, \infty) \text{ model good}$$

FIND b_1 AND b_2 FROM THE TABLE

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	$\beta_0 = 1.3863$	0.5590	2.480	0.01314
x	$\beta_1 = -2.0794$	0.7826	-2.657	0.00788

Null deviance	47.111
Residual deviance	39.112

Terminated	Over40: No	Over40: Yes
Yes	17	71
No	564	835

$$\ln \left(\frac{\mu_i}{1 - \mu_i} \right) = \beta_0 + \beta_1 x_i$$

$$e^{\beta_0 + \beta_1 x_i} = \frac{\mu_i}{1 - \mu_i} \Rightarrow \mu_i = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$$

$$e^{\beta_0 + \beta_1 x_i} = \frac{\mu_i}{1 - \mu_i} \Rightarrow \mu_i = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$$

LIKELIHOOD, LOGLIKELIHOOD AND SCORE FUNCTION

- SCRIVO FORMULA DISTRIBUZIONE
- PRODUTTORIA
- LOGARITMO
- DERIVATA

NORMAL/GAUSSIAN SIMPLE/MULTIPLE LINEAR REGRESSION

$$j(y_i, \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}[y_i - \mu_i]^2\right\} \quad \mu_i = \beta_0 + \beta_1 x_i$$

$$N_i = \beta_0 + \beta_1 x_i$$

likelihood

$$\begin{aligned} L(\theta): &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left\{-\frac{1}{2\sigma^2}[y_i - \mu_i]^2\right\} \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma^2} \prod_{j=1}^n \exp\left\{-\frac{1}{2\sigma^2}[y_i - \mu_j]^2\right\} \\ &= \left(\frac{1}{\sqrt{2\pi}\sigma^2}\right)^n \exp\left\{-\frac{1}{2\sigma^2}\sum_{j=1}^n [y_i - \mu_j]^2\right\} \\ &= \left(\frac{1}{\sqrt{2\pi}\sigma^2}\right)^n \exp\left\{-\frac{1}{2\sigma^2}\sum_{j=1}^n [y_i - \mu_j]^2\right\} \\ &= (2\pi)^{-\frac{n}{2}} (\sigma^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma^2}\sum_{j=1}^n [y_i - \mu_j]^2\right\} \end{aligned}$$

LOGlikelihood

$$\begin{aligned}\chi(\theta) &= \log(L(\theta)) \\ \chi(\theta) &= \log \left\{ (2\pi)^{-\frac{n}{2}} (\sigma^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma^2}} \sum_{i=1}^n [y_i - \mu; -\infty; \infty] \right\} \\ &= \log((2\pi)^{-\frac{n}{2}}) + \log(\sigma^2)^{-\frac{n}{2}} - \frac{1}{2\sigma^2} \sum_{i=1}^n [y_i - \mu; -\infty; \infty] \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n [y_i - \mu; -\infty; \infty]\end{aligned}$$

score function $\lambda_r(\theta)$

DERIVATA DI $l(\theta)$ $\mu: \beta_1 + \beta_2 x;$ 3 VARIABILI $\bullet \beta_1 \bullet \beta_2 \bullet \sigma^2$

$$\begin{aligned}
 \bullet \frac{d}{d\beta_1} l(\theta) &= \frac{d}{d\beta_1} \left\{ -\frac{1}{2} \ln \cos(\pi\beta_1) - \frac{1}{2} \cos(\pi\beta_1) - \frac{1}{2\pi^2} \sum_{i=1}^n [y_i - \beta_1 - \beta_2 x_i]^2 \right\} \\
 &= \frac{1}{2\pi^2} \sum_{i=1}^n \frac{d}{d\beta_1} \left\{ \sum_{i=1}^n [y_i - \beta_1 - \beta_2 x_i] \cdot (-1) \right\} \\
 &= \frac{1}{2\pi^2} \sum_{i=1}^n \left\{ \sum_{i=1}^n [y_i - \beta_1 - \beta_2 x_i] \right\} = \alpha
 \end{aligned}$$

$$\begin{aligned}
 \frac{d}{d\beta_2} \lambda(\theta) &= \frac{d}{d\beta_2} \left\{ -\frac{1}{2} \cos(\pi) - \frac{1}{2} \cos(\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n \{y_i - \beta_0 - \beta_2 x_i\}^2 \right\} \\
 &= \frac{1}{\sigma^2} \frac{d}{d\beta_2} \left\{ \sum_{i=1}^n \{y_i - \beta_0 - \beta_2 x_i\} \cdot \{x_i\} \right\} \\
 &= \frac{1}{\sigma^2} \frac{d}{d\beta_2} \left\{ \sum_{i=1}^n \{y_i - \beta_0 - \beta_2 x_i\} \cdot \{x_i\} \right\} = b
 \end{aligned}$$

$$\begin{aligned}
 \bullet \frac{d}{dx} \ln(\theta) &= \frac{d}{dx} \left[-\frac{1}{2} \ln(\cos^2(x)) - \frac{1}{2} \ln(\cos^2(x)) - \frac{1}{2\sigma^2} \sum_{i=1}^n \left[(y_i - \beta_0 - \beta_1 x_i)^2 \right] \right] \\
 &= -\frac{1}{2} \frac{1}{\cos^2(x)} \cdot \cos(x) \cdot (-\sin(x)) - \frac{d}{dx} \left[\frac{1}{2\sigma^2} \sum_{i=1}^n \left[(y_i - \beta_0 - \beta_1 x_i)^2 \right] \right] \\
 &= -\frac{1}{2} \frac{1}{\cos^2(x)} \cdot \left(\sum_{i=1}^n \left[(y_i - \beta_0 - \beta_1 x_i)^2 \right] \right) \frac{d}{dx} \frac{1}{2\sigma^2} \\
 &= -\frac{1}{2} \frac{1}{\cos^2(x)} \cdot \left(\sum_{i=1}^n \left[(y_i - \beta_0 - \beta_1 x_i)^2 \right] \right) \frac{1}{2\sigma^2} \cdot (-2) \cdot \frac{1}{\cos^2(x)} = C
 \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial \beta_1} \left\{ \sum_{i=1}^n [y_i - \beta_0 - \beta_1 x_i] \right\} &= a \\ \frac{\partial}{\partial \beta_2} \left\{ \sum_{i=1}^n [y_i - \beta_0 - \beta_1 x_i] (x_i) \right\} &= b \\ -\frac{n}{2} \frac{\partial}{\partial \sigma^2} \left\{ -\frac{1}{2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 \right\} \frac{1}{2} \frac{(-1)}{\sigma^2} \left(\frac{1}{\sigma^2} \right) &= c \end{aligned} \Rightarrow c^* = [a, b, c]$$

Score function for multiple linear regression model
basically you will have many b rows in the system that are like this:

$$\frac{1}{2} \frac{d}{d\mu_j} \left\{ \sum_{i=1}^n [y_i - \beta_1 - \beta_2 x_{i1} - \dots - \underbrace{\beta_j x_{ij}}_{\mu_j} - \dots - \beta_n x_{in}] (x_{ij}) \right\}$$

Exact distribution of the estimators $\hat{\beta}$ and $\hat{\beta}_1$

$$\hat{\beta}(y) \sim N_p(\beta; \sigma^2 (X^T X)^{-1})$$

THE MARGINAL $\hat{\beta}_j(\cdot) \sim N\left(\beta_j; \sigma^2 \left[(X^T X)^{-1} \right]_{(j,j)}\right) \quad j=1, \dots, p$

BERNOULLI F, LOGIT/PROBIT=GENERALIZED LINEAR REGRESSION

$$S(y_i, \theta) = \pi_i y_i (1 - \pi_i)^{1-y_i}; \quad \frac{\pi_i}{1 - \pi_i} = e^{\beta_1 + \beta_2 D_i} \rightarrow \pi_i = \frac{e^{\beta_1 + \beta_2 D_i}}{1 + e^{\beta_1 + \beta_2 D_i}} = \frac{e^{\tilde{x}_i^T \beta}}{1 + e^{\tilde{x}_i^T \beta}} \rightarrow \pi_i = \frac{e^{\tilde{x}_i^T \beta}}{1 + e^{\tilde{x}_i^T \beta}}$$

$$\frac{\pi_i}{1-\pi_i} = e^{\beta_1 + \beta_2 D_i} \rightarrow \pi_i = \frac{e^{\beta_1 + \beta_2 D_i}}{1 + e^{\beta_1 + \beta_2 D_i}} = \frac{e^{\tilde{x}_i^T \beta}}{1 + e^{\tilde{x}_i^T \beta}} \rightarrow \pi_i = \frac{e^{\tilde{x}_i^T \theta}}{1 + e^{\tilde{x}_i^T \theta}}$$

likelihood

$$L(\theta) = \prod_i \pi_i y_i (1 - \pi_i)^{1 - y_i} = \prod_i \left(\frac{e^{\alpha_i^T \gamma}}{1 + e^{\alpha_i^T \gamma}} \right)^{y_i} \left(1 - \frac{e^{\alpha_i^T \gamma}}{1 + e^{\alpha_i^T \gamma}} \right)^{1 - y_i}$$

$$= \prod_i \left(\frac{e^{\alpha_i^T \gamma}}{1 + e^{\alpha_i^T \gamma}} \right)^{y_i} \left(\frac{1}{1 + e^{\alpha_i^T \gamma}} \right)^{1 - y_i}$$

LOGlikelihood

[illegible]

score function $\lambda_{\eta}(\beta)$

$$\begin{cases} \frac{d}{d\beta_1} \lambda(\theta) = \sum y_i - \sum \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}} \cdot 1 \\ \frac{d}{d\beta_2} \lambda(\theta) = \sum y_i \cdot x_i - \sum \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}} \cdot x_i \end{cases}$$

$$\lambda_{\beta}(\theta) = \frac{d\lambda(\theta)}{d\theta_{\beta}} = \sum_{i=1}^h \tilde{x}_i^T \theta - \log(1 + e^{\tilde{x}_i^T \theta}) = \sum_{i=1}^n y_i x_{i1} - \frac{1}{1 + e^{\tilde{x}_i^T \theta}} \cdot \tilde{x}_i^T \theta \cdot x_{i1} \\ = \sum_{i=1}^n x_{i1} \left(y_i - \frac{\tilde{x}_i^T \theta}{1 + e^{\tilde{x}_i^T \theta}} \right) = \sum_{i=1}^n x_{i1} (y_i - \pi_i) = \tilde{x}^T (\mathbf{y} - \boldsymbol{\pi})$$

Distribution of the estimators $\hat{\beta}$
(regression parameters)

$$\hat{\underline{\beta}}(\underline{y}) \sim N_p(\underline{\beta}; \tau(\hat{\underline{\theta}})^{-1}) \longleftrightarrow \begin{bmatrix} \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} \sim N_p\left(\begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}; \tau(\hat{\underline{\theta}})^{-1}\right)$$

THE MARGINAL $\hat{\beta}_j$ (?) IN $\left(\beta_j; [x(\hat{\beta})^{-1}]_{(j,j)} \right)$ $j=1, \dots, p$

POISSON=GENERALIZED LINEAR REGRESSION

$$f(y_i; \mu_i) = \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!} \quad \mu_i = e^{\beta_0 + \beta_1 x_i}$$

$$\mu_i = e^{\mathbf{x}_i^T \boldsymbol{\beta}}$$

likelihood

$$L(\theta) = \prod_{i=1}^n \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}$$

LOGlikelihood

$$\begin{aligned} \lambda(\mathbf{y}) &= \text{Log}(\mathcal{L}(\mathbf{y})) = \text{Log} \left\{ \prod_{y_i!} \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!} \right\} = \sum_{i=1}^n \text{Log} \left(\frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!} \right) = \sum_{i=1}^n \text{Log}(e^{-\mu_i}) + \sum_{i=1}^n \text{Log}(\mu_i^{y_i}) - \sum_{i=1}^n \text{Log}(y_i!) \\ &= \sum_{i=1}^n -\mu_i + y_i \text{Log}(\mu_i) - \text{Log}(y_i!) = - \sum_{i=1}^n e^{\sum_{j=1}^p \beta_j x_{ij}} + \sum_{i=1}^n y_i \sum_{j=1}^p \beta_j x_{ij} - \sum_{i=1}^n \text{Log}(y_i!) \end{aligned}$$

score function $J_1(\beta)$

$$\begin{cases} \frac{d}{d\beta_1} \ell(\theta) = -\sum e^{\beta_1 + \beta_2 x_i} \cdot 1 + y_i \\ \frac{d}{d\beta_2} \ell(\theta) = -\sum e^{\beta_1 + \beta_2 x_i} \cdot x_i + y_i x_i \end{cases}$$

$$\frac{d}{d\beta_1} \lambda(\beta) = \frac{d}{d\beta_1} \left[\sum_{i=1}^n e^{\tilde{x}_i \cdot \beta} + \sum_{i=1}^n y_i \tilde{x}_i \cdot \beta - \sum_{i=1}^n \log(y_i!) \right] = \sum_{i=1}^n (e^{\tilde{x}_i \cdot \beta} + x_{i1}) + \sum_{i=1}^n y_i x_{i1} = \sum_{i=1}^n x_{i1} (y_i + e^{\tilde{x}_i \cdot \beta}) = \sum_{i=1}^n x_{i1} (y_i + \lambda_i)$$

Appropriate distribution of the maximum likelihood estimator

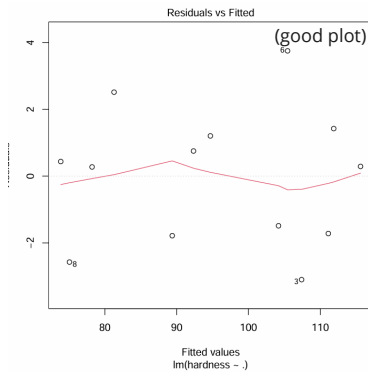
$$\hat{\underline{\beta}}(\underline{y}) \sim N_p(\underline{\beta}; \tau(\hat{\underline{\theta}})^{-1}) \longleftrightarrow \begin{bmatrix} \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} \sim N_p\left(\begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}; \tau(\hat{\underline{\theta}})^{-1}\right)$$

Appropriate distribution of the maximum likelihood estimator $\hat{\beta}_2$ of β_2 (like $\hat{\beta}_1$ of β_1)

THE MARGINAL $\hat{\beta}_j$ IN $\left(\beta_j; [\gamma(\hat{\beta})^{-1}]_{(j,j)} \right) \quad j=1, \dots, p$

GRAPHS

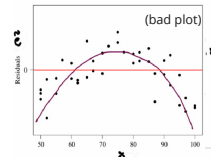
Residuals vs fitted



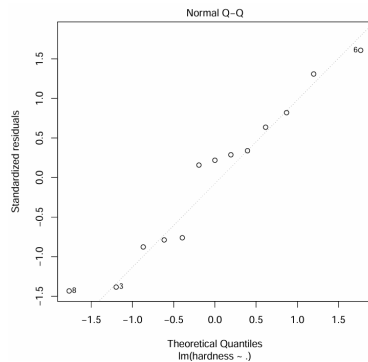
- Use to check if the model's assumptions are satisfied, specifically orthogonality of the residuals and homoscedasticity.
- Good model => plot without particular patterns. No troubling behaviors. No relations.
- Bad model => we can see some patterns of the residuals. Exists a relation
- Not enough observations (n) to make good assumptions

Solutions for bad plot:

- Add other covariates
- Transform covariates
- If the residuals follow a non-linear model add a covariate that resemble the pattern (like x^2)



Normal Q-Q plot

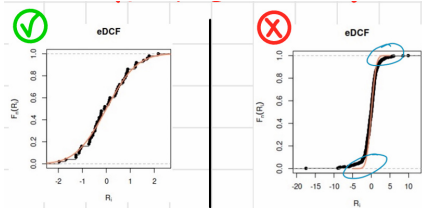


- Use to check the normality assumption: empirical vs theoretical quantiles.
- The plot is used to determine if the residuals of the regression model are normally distributed
- If the hypothesis is satisfied (errors follow a normal distribution) => points should be on the line
- Outliers on the tails
- Not enough observations (n) to make good assumptions

Solution for bad plot:

- If there is a pattern: transform covariates, examine deeply the data, use another model.

eCDF (Empirical Cumulative Distribution Function)

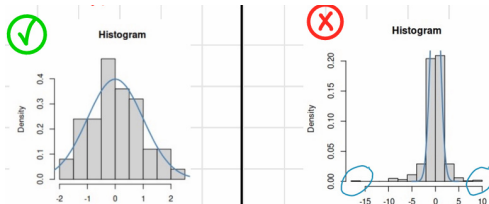


- Plot of studentized residuals vs cumulative proportion of residuals
- X-axis: studentized residual
- Y-axis: cumulative proportion of residuals
- Line: theoretical CDF of $N(0,1)$ (normal standard distribution)
- Plot used to verify if residuals follow a gaussian normal distribution
- Not enough observations (n) to make good assumptions
- Outliers: points very far from the line (tails = extreme data)

Solutions for bad plot:

- If there is a pattern: transform covariates, examine deeply the data, use another model.

Histogram

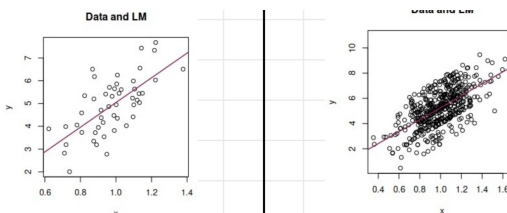


- Plot of studentized residuals vs density of residuals
- X-axis: studentized residual
- Y-axis: density of residuals
- Line: density of $N(0,1)$ (normal standard distribution)
- Bins: interval value. Smaller bins => more details & more precise
- Plot use to verify if residuals follow a gaussian normal distribution
- If bins close to the line => normal distribution
- Not enough observations (n) to make good assumptions
- Outliers: bars very tall or small with respect to the line (tails = extreme data)

Solutions for bad plot:

- If there is a pattern: transform covariates, examine deeply the data, use another model.

Y-X plot



- Plot of independent variables vs response variable Y
- X-axis: X
- Y-axis: Y
- Line: regression line
- Points: value observed
- Not enough observations (n) to make good assumptions
- Plot use to verify if the model is a good model
- If points close to the model => good model

ANOVA

Group	n_j	$\bar{y}_j$	s_j
A ($j = 1$)	4	0.347	0.2055
B ($j = 2$)	3	3.067	0.4618
C ($j = 3$)	8	1.764	0.5007

Sample	Brand A	Brand B	Brand C
1	10.21	11.32	11.60
2	10.25	11.20	11.90
3	10.24	11.40	11.80
4	9.80	10.50	12.30
5	9.77	10.68	12.20
6	9.73	10.90	12.20

- h_j : NUMBER OF OBSERVATION IN GROUP j
- $\bar{y}_j$: MEAN OF GROUP j
- s_j : STANDARD DEVIATION OF GROUP j
- $\bar{y}$: "TOTAL AVERAGE"
- N : TOTAL NUMBER OF OBSERVATIONS $\sum h_j$

$$\bar{y} = \frac{1}{N} \sum h_j \bar{y}_j$$

$$SST = \sum \sum (y_{ij} - \bar{y})^2$$

WITHIN GROUP VARIABILITY

$$SSE = \sum (h_j - 1) s_j^2 = \sum \sum (y_{ij} - \bar{y}_j)^2 = \sum \sum (y_{ij} - \hat{y}_{ij})^2$$

BETWEEN GROUP VARIABILITY

$$SSR = \sum h_j (\bar{y}_j - \bar{y})^2 = \sum \sum (\bar{y}_j - \bar{y})^2 = \sum \sum (\hat{y}_{ij} - \bar{y})^2$$

DEGREES OF FREEDOM (DOF)

$$SST: n-1$$

$$SSE: n-j$$

$$SSR: j-1$$

$$DOF(SST) = DOF(SSE) + DOF(SSR)$$

specify appropriate linear regression model

$$Y_i = \beta_0 + \beta_1 X_{i,2} + \beta_2 X_{i,3} + \varepsilon_i \quad \begin{array}{l} \varepsilon_i: NN(0, \sigma^2) \quad i=1, \dots, N \\ Y_i: NN(\mu_i, \sigma^2) \end{array}$$

$$\begin{cases} Y_i: NN(\beta_0, \sigma^2) & i=1, \dots, 4 \text{ (GROUP A)} \\ Y_i: NN(\beta_0 + \beta_1, \sigma^2) & i=5, \dots, 7 \text{ (GROUP B)} \\ Y_i: NN(\beta_0 + \beta_2, \sigma^2) & i=8, \dots, 15 \text{ (GROUP C)} \end{cases} \quad \begin{array}{l} X_{i,2} = \begin{cases} 1 & \text{IF ANTIDOTE B} \\ 0 & \text{OTHERWISE} \end{cases} \\ X_{i,3} = \begin{cases} 1 & \text{IF ANTIDOTE C} \\ 0 & \text{OTHERWISE} \end{cases} \end{array}$$

$$\begin{aligned} \hat{\beta}_0 &= \mu_A = \bar{y}_A \\ \hat{\beta}_0 + \hat{\beta}_1 &= \mu_B = \bar{y}_B \longrightarrow \hat{\beta}_1 = \bar{y}_B - \bar{y}_A \\ \hat{\beta}_0 + \hat{\beta}_2 &= \mu_C = \bar{y}_C \longrightarrow \hat{\beta}_2 = \bar{y}_C - \bar{y}_A \end{aligned}$$

hyp. testing on average

$$\begin{cases} H_0: \mu_1 = \mu_2 = \mu_3 \\ H_1: \neg H_0 \end{cases}$$

$$f^{OBS} \sim F_{j-1; n-j}$$

$$f^{OBS} = \frac{\frac{SSR}{j-1}}{\frac{SSE}{n-j}}$$

$$\alpha^{OBS} = P(F_{j-1; n-j} > f_{OBS}) \approx 1 - \alpha \approx \dots$$